Bayesian Network Reconstruction of Interbank Exposures (Nepal)

Krishna Neupane

2026-10-17

Abstract

text text

1 Paper 1 — Bayesian Network Reconstruction of Interbank Exposures (Nepal)

Status: design settled, not yet implemented. **Target venue:** ICLR / NeurIPS / ICAIF, 7 pages.

One-line description: Reconstruct the latent bilateral interbank exposure matrix W_t for Nepali BFIs from NRB bulletin data, using an SBM-structured, covariate-informed generative model with a differentiable regulatory-constraint projection layer, amortized by a GNN/GRU inference network.

This supersedes the earlier version of `feature_model_mapping.md`, which had drifted from the actual design (it invented a generic "Bayesian-net structural-constraint" layer built from accounting identities — that was never part of the real design and is corrected below).

/home/krishna/Documents/personal/neupane_article/network_papers/00_llama_arch/muse_llama_arch/apps/open-nrb

1.1 0. Problem framing and contribution to policy

Problem framing (what "the network" means here). We do not pre-specify the edges of the interbank network. Instead, following the network reconstruction literature (e.g., Squartini et al., 2018), we treat the bilateral exposure structure as latent and infer it from publicly observable aggregates — balance-sheet marginals and bank-level covariates — within a constrained statistical framework. Concretely: a node is a bank; a directed edge $i \to j$ is $W_{ij,t}$, the exposure of bank i to bank j; an edge "exists" when $W_{ij,t} > 0$. Neither the edge set nor the weights are observed — only the row/column marginals and covariates are. This is a problem of statistical inference under partial observability, not analysis over a topology that is already known or self-reported. (Revisit/tighten this paragraph later — current form is a first pass.)

Contribution to policy. No central bank in a resource-constrained setting is going to mandate new bilateral exposure reporting purely for a research exercise — that is a costly, politically slow regulatory

change. The contribution here is that this doesn't require that change: a usable, continuously-updatable estimate of systemic interconnectedness is produced from data NRB (or any similarly-resourced central bank) already publishes every month, with zero new reporting burden. Put to a central banker directly: today there is no way to ask "who is exposed to whom" or "which banks are structurally systemic" — not because the question doesn't matter, but because answering it was assumed to require data that isn't collected. This method shows it doesn't. It converts a previously unanswerable supervisory question into an answerable one, cheaply, and produces a reusable pipeline that other analyses (stress tests, contagion simulations, Paper 2's spillover estimation) can plug into. This also generalizes beyond Nepal: most emerging-market central banks publish the same kind of aggregate call-report data and face the identical constraint, so the method is a template for resource-constrained regulators broadly, not a Nepal-specific workaround.

1.2 1. Why this is its own paper

Separated from the causal/policy question (Paper 2) because the two are methodologically unrelated: this paper's contribution is a reconstruction method with no causal-identification content; Paper 2's contribution is a causal-identification design that *consumes* this paper's output as a fixed input. Bundling them would force a methods paper to carry a policy narrative it doesn't need, and vice versa.

1.3 2. Notation

Symbol	Meaning
$i, j \in \{1, ..., N\}$	institutions (BFIs)
$t \in \{1, ..., T\}$	periods (NRB bulletin months)
$X_{i,t} \in \mathbb{R}^d$	node covariates for bank i at time t
$r_{i,t}, c_{i,t} \geq 0$	observed interbank lending / borrowing marginals
$K_{i,t}$	core capital (used to set the regulatory single-obligor cap)
$W_t \in \mathbb{R}_{\geq 0}^{N \times N}$	**latent** bilateral exposure matrix — never observed, the reconstruction target
$z_i \in \{1, ..., K\}$	latent block/tier membership (e.g. core vs. periphery)

1.4 3. Generative model

- Block membership: $z_i \sim \text{Categorical}(\pi)$ — (Eq. 1)
- Block-pair intensity: $\Lambda \in \mathbb{R}_{\geq 0}^{K \times K}$ — the SBM core
- Fitness/degree-correction from covariates: $\phi_{i,t} = \exp(w^\top X_{i,t})$ — (Eq. 2)

- Latent edge intensity:

$$\lambda_{ij,t} = \Lambda_{z_i z_j} \cdot \phi_{i,t}\phi_{j,t} \cdot \exp(h_{i,t}^\top h_{j,t}) \qquad \text{(Eq. 3)}$$

- Zero-inflated edge weight (most bank pairs have *no* exposure):

$$\Pr(\tilde{W}_{ij,t} = 0) = \sigma(-\eta_{ij,t}), \qquad \tilde{W}_{ij,t} \mid \tilde{W}_{ij,t} > 0 \sim \text{LogNormal}(\log \lambda_{ij,t}, \sigma^2) \qquad \text{(Eq. 4)}$$

- Marginals are a **deterministic function** of W_t:

$$r_{i,t} = \sum_j W_{ij,t}, \qquad c_{j,t} = \sum_i W_{ij,t} \qquad \text{(Eq. 5)}$$

  This means the marginal constraint belongs in the *architecture* (a hard projection), not the likelihood — see §5.

1.5 4. Inference network (amortized, PyG GNN + GRU)

The encoder runs on covariates/history only — it cannot message-pass over W_t itself, since that would be circular (the whole point is that W_t is unobserved).

- Temporal encoder: $h_{i,t} = \text{GRU}_\theta(X_{i,t}, h_{i,t-1})$ (Eq. 6), optionally refined by a GNN layer over an auxiliary *observed* graph if one exists (e.g. common-sector exposure — see §8).
- Amortized block-assignment posterior: $q_\psi(z_i \mid X_{i,1:t}) = \text{Categorical}(\text{softmax}(\text{MLP}_\psi(h_{i,t})))$ (Eq. 7)
- Pairwise decoder produces $(\eta_{ij,t}, \lambda_{ij,t})$ from $(h_{i,t}, h_{j,t}, z_i, z_j)$ — full amortized variational posterior $q_\psi(W_t \mid X_{1:t}, r_t, c_t)$
- Reparameterized sampling: log-normal reparam trick for the continuous weight, Gumbel-softmax for the discrete existence gate.

1.6 5. Constraint-satisfaction layer (the technical core of the paper)

Differentiable **capacitated Sinkhorn / entropic-OT projection**:

$$W_t = \text{Sinkhorn}_{\text{cap}}(\tilde{W}_t;\ r_t, c_t, \kappa K_t) \qquad \text{(Eq. 8)}$$

Enforces, exactly and differentiably: 1. row/column marginals match the observed $r_{i,t}, c_{i,t}$ 2. the regulatory single-obligor cap $W_{ij,t} \le \kappa K_{i,t}$

References: Cuturi (2013) entropic OT; Mena et al. (2018) Gumbel-Sinkhorn. Flagged explicitly as "the one genuinely nontrivial implementation piece" in the design discussion — this is the part most likely to need real engineering effort, not the GNN/GRU scaffolding around it.

Correction vs. the earlier `feature_model_mapping.md`: there is no "Bayesian-net structural-constraint edge" tying `TOTAL Liabilities = TOTAL ASSETS` or `Net Profit` into the graph. That was

never part of this design. The accounting identities in the raw NRB statements are a **data-quality cross-check** on the extractor (do the reported totals reconcile?), not a component of the generative model or the constraint layer.

1.7 6. Training objective

$$\mathcal{L} = -\mathbb{E}_q[\log p(W_t^{\text{true}})]_{\text{synthetic only}} + \mathrm{KL}(q_\psi(z) \,\|\, p(z)) + \lambda_{\text{smooth}} \sum_t \|W_t - W_{t-1}\|_F^2 + \lambda_{\text{reg}} \|\Lambda, w\|^2$$

- **Synthetic data** (known ground-truth W_t from an SBM generator): the first term is a real supervised reconstruction loss — this is the main validation evidence.
- **Real data** (no W_t is ever observed): the first term drops out entirely. What's left is a regularized/MAP-style constrained fit, *not* likelihood-based estimation. Evaluation on real data has to rely on indirect evidence: temporal stability of the recovered W_t sequence, and comparison against a classical baseline (§8).

1.8 7. Design rationale

- **SBM over preferential-attachment/fitness models** as the sole structural backbone: core-periphery tiering in interbank markets is literally a latent-block story, so SBM is the natural fit. PA/fitness models (Caldarelli et al.) are cited in related work for contrast but not built.
- **GRU, not a temporal-GNN** (EvolveGCN, TGN, etc.) for the temporal mechanism: deliberately the simpler choice, given the "big model, small data" risk — roughly 50–80 institutions, at most ~170 periods. A fancier temporal-GNN would be over-parameterized for this panel size.
- **Mandatory baseline**: head-to-head comparison against Gandy–Veraart / RAS maximum-entropy network reconstruction, on both the synthetic SBM data (known ground truth) and the real panel. Explicit instruction from the design discussion: report honestly if the deep model doesn't beat the classical baseline — don't bury a negative result.

1.9 11. Algorithms (pseudocode)

Three algorithms, in dependency order: the capacitated Sinkhorn projection (the hard constraint layer, Eq. 8) is standalone and testable without the rest of the model; the generative forward pass (Eq. 1–5, 8) samples a synthetic W_t for validation; training (Eq. 6–9) wraps the amortized encoder/decoder of §4 around the same projection layer. All three are written over *symbol-level* inputs (N, K, d, κ, etc.) — nothing below is a fixed dataset size or constant; see §12 for how these get threaded through as config, not literals.

1.10 8. Variable inventory (what actually feeds the model)

Net statement from the design discussion: **the reconstruction model needs ~5 covariates per node per period, plus the two interbank marginals. Everything else in the NRB bulletins is either a Paper 2 outcome variable or not usable given current data coverage.**

Algorithm 1 CapacitatedSinkhornProjection (Eq. 8)

Require: $\tilde{W} \in \mathbb{R}_{\geq 0}^{N \times N}$ (raw/sampled edge weights), $r, c \in \mathbb{R}_{\geq 0}^{N}$ (target marginals), $\text{cap} \in \mathbb{R}_{\geq 0}^{N}$ (per-row cap, $\text{cap}_i = \kappa \cdot K_{i,t}$), $\varepsilon > 0$ (entropic regularization), $n_{\text{iters}}, \text{tol}$ (stopping criteria)

Ensure: $W \in \mathbb{R}_{\geq 0}^{N \times N}$ satisfying row/col marginals and $W_{ij} \leq \text{cap}_i$ exactly

$\text{Kern} \leftarrow \exp(\log(\tilde{W} + \delta)/\varepsilon)$ ▷ δ tiny avoids log(0); config, not literal
$u \leftarrow \mathbf{1}_N$; $v \leftarrow \mathbf{1}_N$
for iter $= 1 \ldots n_{\text{iters}}$ **do**
  $u_{\text{new}} \leftarrow r/(\text{Kern} \cdot v)$
  $u_{\text{new}} \leftarrow \min(u_{\text{new}}, \text{cap}/(\text{Kern} \cdot v))$ ▷ Enforce row-wise cap
  $v_{\text{new}} \leftarrow c/(\text{Kern}^T \cdot u_{\text{new}})$
  if $\|u_{\text{new}} - u\| < \text{tol}$ **and** $\|v_{\text{new}} - v\| < \text{tol}$ **then**
    break
  end if
  $u, v \leftarrow u_{\text{new}}, v_{\text{new}}$
end for
$W \leftarrow \text{diag}(u) \cdot \text{Kern} \cdot \text{diag}(v)$
return W

Algorithm 2 SampleSyntheticNetwork (Eq. 1–5, 8; ground truth for validation, §6)

Require: N, K, d (problem dimensions), $\pi \in \Delta^K, \Lambda \in \mathbb{R}_{\geq 0}^{K \times K}, w \in \mathbb{R}^d, \sigma^2$ (generative params), $X_{1:N} \in \mathbb{R}^{N \times d}$ (covariates), $\kappa, K_{\text{cap}} \in \mathbb{R}^N$ (regulatory cap inputs)

Ensure: $W_{\text{true}} \in \mathbb{R}_{\geq 0}^{N \times N}, r, c \in \mathbb{R}^N$ (ground-truth network + marginals)

for $i = 1 \ldots N$ **do**
  $z_i \sim \text{Categorical}(\pi)$ ▷ Eq. 1
end for
for $i = 1 \ldots N$ **do**
  $\varphi_i \leftarrow \exp(w^T X_i)$ ▷ Eq. 2
end for
for $i, j = 1 \ldots N$ $(i \neq j)$ **do**
  $\lambda_{ij} \leftarrow \Lambda[z_i, z_j] \cdot \varphi_i \cdot \varphi_j$ ▷ Eq. 3
  $e_{ij} \sim \text{Bernoulli}(\sigma(\eta_{ij}))$ ▷ Eq. 4, zero-inflation param
  $\tilde{W}_{ij} \leftarrow e_{ij} \cdot \text{LogNormalSample}(\log \lambda_{ij}, \sigma^2)$
end for
$r \leftarrow \text{row_sums}(\tilde{W})$; $c \leftarrow \text{col_sums}(\tilde{W})$
$W_{\text{true}} \leftarrow \text{CapacitatedSinkhornProjection}(\tilde{W}, r, c, \kappa \cdot K_{\text{cap}})$ ▷ Eq. 8
return $W_{\text{true}}, \text{row_sums}(W_{\text{true}}), \text{col_sums}(W_{\text{true}})$

Algorithm 3 TrainStep (Eq. 6–9; amortized inference + one optimizer step)

Require: Batch of periods $\{X_{1:N,t}, r_t, c_t, K_t\}_{t \in \text{batch}}$, optional $W_{\text{true},t}$, model params $\theta, \psi, \Lambda, w, \sigma^2$, hyperparams $\lambda_{\text{smooth}}, \lambda_{\text{reg}}, \kappa$
State: $h_{i,t-1}$ carried across periods (GRU hidden state)
Ensure: Loss value; updated θ, ψ, Λ, w after one gradient step

for $t \in$ batch (temporal order) **do**
  for $i = 1 \dots N$ **do**
    $h_{i,t} \leftarrow \text{GRU}_\theta(X_{i,t}, h_{i,t-1})$ ▷ Eq. 6
    [Optional] $h_{i,t} \leftarrow \text{GNN_refine}(h_{i,t}, A_{\text{obs},t})$ ▷ §4, §8c
    $q_i \leftarrow \text{softmax}(\text{MLP}_\psi(h_{i,t}))$ ▷ Eq. 7
    $z_i \sim \text{GumbelSoftmaxSample}(q_i)$ ▷ Reparameterized
  end for
  for $i, j = 1 \dots N$ $(i \neq j)$ **do**
    $(\eta_{ij,t}, \lambda_{ij,t}) \leftarrow \text{Decoder}_\psi(h_{i,t}, h_{j,t}, z_i, z_j)$ ▷ Eq. 3–4
    $e_{ij,t} \sim \text{GumbelSigmoidSample}(\eta_{ij,t})$ ▷ Reparameterized gate
    $\tilde{W}_{ij,t} \leftarrow e_{ij,t} \cdot \text{LogNormalSample}(\log \lambda_{ij,t}, \sigma^2)$
  end for
  $W_t \leftarrow \text{CapacitatedSinkhornProjection}(\tilde{W}_t, r_t, c_t, \kappa \cdot K_t)$ ▷ Eq. 8, Alg. 1
  if $W_{\text{true},t}$ is given (synthetic branch) **then**
    $L_{\text{recon},t} \leftarrow -\log p(W_{\text{true},t} \mid \lambda_t, \eta_t, \sigma^2)$ ▷ Eq. 9, first term
  else
    $L_{\text{recon},t} \leftarrow 0$ ▷ Real data: no ground truth
  end if
  $L_{\text{KL},t} \leftarrow \text{KL}(q_\psi(z_t) \,\|\, p(z))$
end for
$L_{\text{smooth}} \leftarrow \sum_t \|W_t - W_{t-1}\|_F^2$ ▷ Over consecutive periods in batch
$L_{\text{reg}} \leftarrow \|\Lambda\|^2 + \|w\|^2$
$L \leftarrow \sum_t (L_{\text{recon},t} + L_{\text{KL},t}) + \lambda_{\text{smooth}} \cdot L_{\text{smooth}} + \lambda_{\text{reg}} \cdot L_{\text{reg}}$
Backpropagate L; update θ, ψ, Λ, w via optimizer step
return L

1.10.1 8a. The two marginals (non-negotiable — the reconstruction target)

NRB field	Statement / side	Symbol
`2.b Interbank Borrowing`	Balance Sheet, **Liabilities**	$c_{i,t}$ (column marginal)
`INTERBANK LENDING (Marginal r_i)`	Balance Sheet, **Assets**	$r_{i,t}$ (row marginal)

These are the actual payoff variables for the whole method — everything else exists to help infer the matrix whose row/column sums these fix.

1.10.2 8b. Node covariates $X_{i,t}$ (block-membership / fitness prior)

#	Covariate	Built from	Role
1	Size	sum of the 11 `ASSET_TOP_LEVEL` columns: `1 LIQUID FUNDS`, `2 INVESTMENT IN SECURITIES`, `3 SHARE & OTHER INVESTMENT`, `4 LOANS & ADVANCES`, `5 BILL PURCHASED`, `6 LOANS AGAINST COLLECTED BILLS`, `7 FIXED ASSETS`, `8 OTHER ASSETS`, `9 EXPENSES NOT WRITTEN OFF`, `10 NON BANKING ASSETS`, `INTERBANK LENDING (Marginal r_i)` (**not** the `TOTAL ASSETS` label directly — see data caveat below)	SBM block/fitness prior, scale control
2	Capital ratio	`1 CAPITAL FUND` (Liabilities) ÷ size	solvency/fitness term; also sets the obligor cap $K_{i,t}$
3	Liquidity ratio	`1 LIQUID FUNDS` (Assets) ÷ size	liquidity position

#	Covariate	Built from	Role
4	Loan-to-deposit pressure	`4 LOANS & ADVANCES` (Assets) ÷ `3 DEPOSITS` (Liabilities)	funding-pressure signal
5	NRB-dependence	`2.a Borrowings - NRB` ÷ `2 BORROWINGS` (both Liabilities; if `2 BORROWINGS` is missing for a row — some early-era filings never print the top-level subtotal — the denominator is reconstructed as the sum of its sub-items: `2.a Borrowings - NRB`, `2.b Interbank Borrowing`, `2.c Foreign Banks and Financial Institutions`, `2.d Other Financial Institutions`, `2.e Bonds and Securities / NRB Bonds`, `2.f Borrowings - Govt Non-Financial Institutions`, `2.g Borrowings - Other Non-Financial Institutions`)	central-bank liquidity dependence — useful for the zero-inflation gate (a bank funded mainly by NRB is less likely to have interbank exposure)

Optional secondary signal, not in the core 5: ownership type (state/private/foreign), if available — would add homophily structure to the block prior.

1.10.3 8c. Auxiliary observed-graph candidate (optional GNN refinement input, §4)

Source	Role
Statement of Credit, sector-wise breakdown (12 sectors + Others)	shared sector-concentration = a correlated-shock signal independent of the interbank marginals. Candidate as the auxiliary *observed* graph the GNN layer can refine over, since it doesn't require W_t itself.

1.10.4 8d. Explicitly excluded from Paper 1

- `6 Bills Payable`, `5 Other Liabilities` (Liabilities, minor residuals) — not used
- `6/10 Reconciliation A/c` (both sides) — accounting plug, not economically meaningful
- `8 Expenses not Written off`, `9 Non Banking Assets` (Assets) — legacy/minor, not used
- Everything in Profit & Loss — these are Paper 2 outcome variables, not reconstruction inputs (see `paper2_causal_spillover.md`)
- `nrb_statement_of_credit.csv` as a panel source — only 2 years of coverage in current data, can't support a 14-year panel; not worth cleaning further unless more years turn up uncollected

1.10.5 8e. One-to-one equation-symbol raw-variable map

Every symbol that appears in Eq. (1)–(8) is listed below with exactly what real NRB field (if any) it is computed from. Symbols with no raw field are latent or learned — they're estimated by the model, not read off the bulletin.

Symbol	Appears in	Raw NRB field	Statement / side	Transformation
$r_{i,t}$	Eq. (5), (8)	`INTERBANK LENDING (Marginal r_i)`	Balance Sheet, Assets	identity (reported value)
$c_{i,t}$	Eq. (5), (8)	`2.b Interbank Borrowing`	Balance Sheet, Liabilities	identity (reported value)
$K_{i,t}$	Eq. (8), $\kappa K_{i,t}$ cap	`1 CAPITAL FUND`	Balance Sheet, Liabilities	identity (reported value) — note: raw value here, *not* the ratio used in $X^{(2)}_{i,t}$ below
$X^{(1)}_{i,t}$	Eq. (2), component of $X_{i,t}$	sum of the 11 `ASSET_TOP_LEVEL` columns (listed in §8b)	Balance Sheet, Assets	sum (not the `TOTAL ASSETS` label — see §9.1)

Symbol	Appears in	Raw NRB field	Statement / side	Transformation
$X^{(2)}_{i,t}$	Eq. (2)	`1 CAPITAL FUND`	Balance Sheet, Liabilities	$\div X^{(1)}_{i,t}$ (capital ratio)
$X^{(3)}_{i,t}$	Eq. (2)	`1 LIQUID FUNDS`	Balance Sheet, Assets	$\div X^{(1)}_{i,t}$ (liquidity ratio)
$X^{(4)}_{i,t}$	Eq. (2)	`4 LOANS & ADVANCES` ÷ `3 DEPOSITS`	Balance Sheet, Assets ÷ Liabilities	ratio across statement sides
$X^{(5)}_{i,t}$	Eq. (2)	`2.a Borrowings - NRB` ÷ `2 BORROWINGS`	Balance Sheet, Liabilities (both)	ratio within Liabilities; denominator falls back to the sum of `2.a`–`2.g` sub-items when `2 BORROWINGS` is unreported (§8b)
w	Eq. (2)	—	—	**learned** (fitness weight vector, dim = 5, no raw field)
z_i	Eq. (1), (3), (7)	—	—	**latent** (inferred via q_ψ, Eq. 7); optionally informed by ownership type if used (§8b)
Λ	Eq. (3)	—	—	**learned** ($K \times K$ block-pair intensity matrix)
$h_{i,t}$	Eq. (3), (6), (7)	$X^{(1\text{-}5)}_{i,t}$ (all 5 covariates, fed in as the GRU input vector)	—	**learned embedding**, GRU output — not read directly off any single field
θ (GRU weights)	Eq. (6)	—	—	**learned**
ψ (posterior-net weights)	Eq. (7)	—	—	**learned**

Symbol	Appears in	Raw NRB field	Statement / side	Transformation
$\eta_{ij,t}$	Eq. (4)	—	—	**learned** (zero-inflation logit, decoder output)
σ^2	Eq. (4)	—	—	**learned** (log-normal dispersion — global or per-block, §10)
κ	Eq. (8)	—	—	**exogenous regulatory constant** (single-obligor exposure limit; not reported in the bulletin itself — a known regulatory parameter)
A^{obs} (auxiliary GNN graph, §4)	Eq. (6) refinement, optional	sector-wise credit, 12 sectors	Statement of Credit	co-exposure adjacency built from shared-sector lending shares

Reading guide: $X_{i,t} = (X_{i,t}^{(1)}, ..., X_{i,t}^{(5)}) \in \mathbb{R}^5$ is the concrete instantiation of the generic $X_{i,t} \in \mathbb{R}^d$ in §2 — i.e. $d = 5$ for this model, and the ordering above (size, capital ratio, liquidity ratio, LDR, NRB-dependence) is the fixed component order used everywhere $X_{i,t}$ appears in Eq. (2) and (6).

1.11 9. Known data-pipeline caveats (apply before this goes into the model)

1. **`item_code` collision**: confirmed by re-running `nrb_extractor.py` against a real bulletin — `item_code` is not unique across the Balance Sheet's Liabilities and Assets sides. Both sides independently restart numbering at 1 (e.g. `1 CAPITAL FUND` on Liabilities vs. `1 LIQUID FUNDS` on Assets; same collision pattern for `2`, `3`, and lettered sub-items `a.`–`g.`). The extractor (`nrb_extractor.py:197-269`, `_extract_rows`) currently has no column marking which side a row is on — that currently has to be inferred externally from which `TOTAL` block (`TOTAL Liabilities` vs. `TOTAL ASSETS`) a row falls between on the page. **Fix needed**: add a `side` column to the extractor output before any join on `item_code` alone, or it will silently collide

liabilities and assets rows that share a code (e.g. covariates 2 and 3 above, which both use an `item_code == 1` row from different sides).

2. Rotated bank-name header recovery (`recover_bank_names`, `nrb_extractor.py:130-171`) and the layout-mode row extraction are both working as validated against the real sample PDF — no open bug there as of this pass.

1.12 10. Open items (not yet decided)

- K **(number of SBM blocks)**: start with $K = 2$ (core/periphery), treat larger K as a robustness check, not the headline specification.
- σ^2 **(log-normal dispersion)**: learned, either globally or per-block — not yet decided which.
- **Venue/scope hook**: ICAIF fit is comfortable at the current scope (methods + real-data case study). ICLR/NeurIPS main track would need an added theoretical hook — e.g. a provable statement on how reconstruction uncertainty in $\hat{W}_t$ propagates into coverage guarantees for a downstream interference estimator (which would also strengthen the Paper 1 → Paper 2 link). **Not resolved** — needs a decision before drafting.

1.13 11. Paper structure (7 pages)

Section	Length	Content
Intro + related work	~1.5p	position against Aronow-Samii/Leung (network assumed *known*), Elliott-Golub-Jackson (network assumed known, no reconstruction), Gandy-Veraart (reconstruction without covariate-informed priors or regulatory constraints)
Method	~2.5p	§3–6 above
Synthetic validation	~1.5p	SBM-generated ground truth, baseline comparison
Real-data case study	~1p	Nepal panel, temporal-stability evidence
Discussion/limitations	~0.5p	honest treatment of the no-ground-truth-on-real-data limitation